

Cellular Automata, $(\mu + \lambda)$ Evolutionary Strategies and Morphogenesis: A CBR Approach

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Abstract. Modeling complex biological systems such as morphogenetic processes requires finding local interaction rules that produce coherent global patterns from strictly local information. In this paper we present a framework that couples *cellular automata* (CA) with a $(\mu + \lambda)$ *evolutionary strategy* $((\mu + \lambda)$ -ES) to automatically discover transition rules that reproduce target morphogenetic patterns, ranging from classical 1D benchmarks to 2D flag-generation problems. Our key contribution is the integration of this optimisation engine into a *Case-Based Reasoning* (CBR) system: new target cases are compared and evaluated exclusively against an initial case base consisting of previously solved foundational patterns (Japan and Hungary). Evolved rules are reused to warm-start the GA for unseen problems, dramatically reducing convergence time. We evaluate the fitness function across three complementary similarity metrics (SSIM, Jaccard index, Mutual Information) and report their individual and combined impact on rule quality. Experimental results show that the CBR engine reduces the number of generations required by up to 80% while maintaining competitive fitness scores.

Keywords: Cellular automata · Genetic algorithms · Case-based reasoning · Morphogenesis · $\mu + \lambda$ strategy · Emergent computation · Similarity metrics

1 Introduction

Morphogenesis, the process by which biological organisms develop their spatial structure from a uniform mass of cells, has fascinated scientists since Turing’s seminal work on reaction-diffusion systems [9,16]. At its core, morphogenesis is governed by *local* biochemical interactions that produce striking *global* order — a phenomenon that computational models struggle to replicate efficiently.

Cellular Automata (CA) are discrete dynamical systems that serve as a natural modeling tool for these processes. In a CA, each cell updates its state according to a fixed local rule depending only on its neighbors [10]. If these underlying rules are successfully found, a deterministic and explainable behavior can be assigned to complex dynamic phenomena whose evolution is, a priori, highly

unpredictable. Consequently, this framework does not simply aim to transform a random input pattern into a target output, but rather to explicitly discover the specific logic and local interactions that drive a pattern’s transformation.

The central challenge lies in identifying the optimal CA transition rule within an exponentially large search space. For instance, a 2D binary CA with a Moore neighborhood yields 2^{512} candidate rules, making manual or brute-force search impossible. Genetic Algorithms (GAs) are well-suited for this task due to their ability to evaluate enormous search spaces and find high-quality solutions [11,5].

However, a practical limitation of GA-based search is the “utility problem”: each new morphogenetic target requires a full optimisation run from scratch, discarding all previously acquired expert knowledge. Following biological parallelism, where evolution reuses successful genetic structures [9], this work proposes integrating CA and an evolutionary strategy optimisation within a Case-Based Reasoning (CBR) engine [1]. This architecture provides the system with a memory to leverage prior experience, evaluating problem similarity against a foundational case base to automate configuration and perform transfer learning by injecting successful rules as initial seeds [3].

The main contributions of this paper are:

1. **Implementation of a $\mu + \lambda$ Evolutionary Strategy:** We propose an adaptive mutation strategy to optimise local transition rules, successfully overcoming epistasis and premature convergence commonly found in classical GAs [14].
2. **Design of a 2D Evaluation Framework:** We introduce an advanced framework for artificial morphogenesis by combining structural (multichannel RGB SSIM), spatial (weighted Jaccard Index), and statistical dependence (Mutual Information) metrics [13].
3. **Development of a CBR Engine:** We present a CBR architecture that automates system parameterization featuring topological retrieval against a dedicated 2D case base (Japan and Hungary) and transfer learning of CA rules [3,18], demonstrating a reduction in computational cost of up to 80%.
4. **Systematic Evaluation:** A comprehensive validation on 1D benchmarks (Density Classification, French Flag Problem, *Drosophila* embryogenesis) and 2D flag-generation problems.

To ensure reproducibility, the full implementation is publicly available at https://github.com/sergiio8/TFG_CS, alongside an interactive web interface to properly visualize the 2D spatiotemporal evolutions <https://gaia.fdi.ucm.es/research/cellautomata/>.

The rest of the paper is organised as follows. Section 2 surveys related work. Section 3 describes the optimisation framework and 2D metrics. Section 4 details the CBR architecture. Section 5 reports experimental results, and Section 6 concludes the paper.

2 Related Work

2.1 Evolving Cellular Automata with Genetic Algorithms

Identifying the optimal CA transition rule to approximate a desired target pattern represents a massive combinatorial challenge. Typically, the transition rule is encoded as a vector where each position represents a possible configuration of a cell and its neighborhood, and the value dictates the new state of the central cell.

For a 2D binary CA with a von Neumann neighborhood ($N = 4$), the rule table has $2^5 = 32$ entries, yielding 2^{32} candidate rules. This vast search space renders exhaustive search methods infeasible, making GAs particularly well-suited to handle the inherent epistasis of rule tables robustly.

Mitchell et al. [11] provided the first systematic study of GA-evolved CA rules on the Density Classification Problem (DCP), achieving around 82% accuracy on random 149-cell automata. The Gacs–Kurdyumov–Levin (GKL) rule [6], designed by hand, has long served as a reference with approximately 96–98% accuracy at larger sizes.

For 2D shape generation, Mouret et al. [13] demonstrated GA-evolved CAs capable of generating complex geometric shapes, reporting promising results when using fitness functions based on mutual information and marginal entropies.

2.2 Case-Based Reasoning in Optimisation

CBR is a problem-solving paradigm in which new problems are solved by adapting solutions to similar past cases [1]. In the context of evolutionary computation, recent works have successfully applied CBR for the automatic parameter tuning of GAs. For instance, E. Yeguas et al. proposed a Bayesian CBR system that utilises probabilistic models to determine the optimal configuration of evolutionary algorithms [18]. Furthermore, transfer learning within evolutionary frameworks has been explored through selective fine-tuning, where knowledge from previously solved optimisation tasks is reused to bias the initial search towards promising regions [3]. These works provide a foundational justification for our approach of warm-starting the $(\mu + \lambda)$ -ES by injecting rules from similar morphogenetic cases.

2.3 Neural Cellular Automata (NCA)

As a prominent alternative, Neural Cellular Automata [12] replace the discrete look-up table with a small convolutional network, enabling continuous-state automata that handle complex textures and regenerative patterns. However, NCA optimize convolutional weights rather than explicit transition rules, making them ill-suited for studying emergent computation: the goal of this work is not to map an input pattern to an output, but to discover the specific local interaction rules that give rise to a target morphogenetic process. This requirement for interpretability and exact rule extraction motivates our choice of deterministic CA over NCA, despite the latter’s greater expressive power.

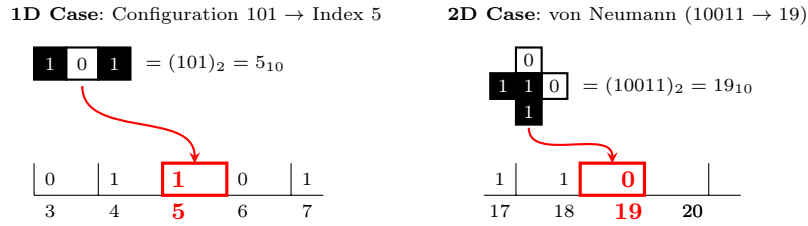


Fig. 1: Mapping logic to retrieve next state from the chromosome

3 Design of the Optimisation Framework

3.1 Cellular Automaton Modeling

To implement the (CA), we utilised the Python library CellPyLib [2]. The transition rules are encoded as a vector where each element dictates the central cell's new state. Assuming M states and a neighborhood of N_{cells} , local configurations are mapped to a base-10 index, interpreting the cell states as a base- M number. Consequently, the search space consists of M^{N+1} possible transition rules, which renders manual search impossible.

Key decisions regarding CA parameterization depend on the problem's topology:

- **Neighborhood selection (2D topologies):** For binary states, we utilise a Moore neighborhood ($N = 8$). However, for problems with three or more states, a von Neumann neighborhood ($N = 4$) is preferred to prevent exponential growth of the chromosome (e.g., $3^5 = 243$ genes vs. $3^9 = 19683$).
- **Boundary handling:** By default, CellPyLib uses periodic boundaries, suitable when opposite edges share the same state. However, we also implemented fixed boundaries for specific rows or columns, providing the minimal constraints necessary for the automaton to stabilize asymmetric patterns.

3.2 Evolutionary Strategy and Adaptive Mutation

We adopted the $(\mu + \lambda)$ -ES strategy [14] —implemented via the DEAP library— to mitigate epistasis and premature convergence. In this approach, λ offspring are generated exclusively through mutation from the current population μ . The next generation is then formed by selecting the best μ individuals from the combined union of parents and offspring. Finally, a 2% diversity mechanism replaces the least fit individuals with random genomes to prevent stagnation and escape local optima.

We further implemented an *adaptive mutation schedule* where the number of mutated genes per offspring, $m(f)$, decreases as a step function of the best individual's fitness f to balance early exploration with fine-grained exploitation near convergence. For complex 2D scenarios —specifically, 2-state automata with a Moore neighborhood yielding a chromosome length of $L = 2^9 = 512$ genes—

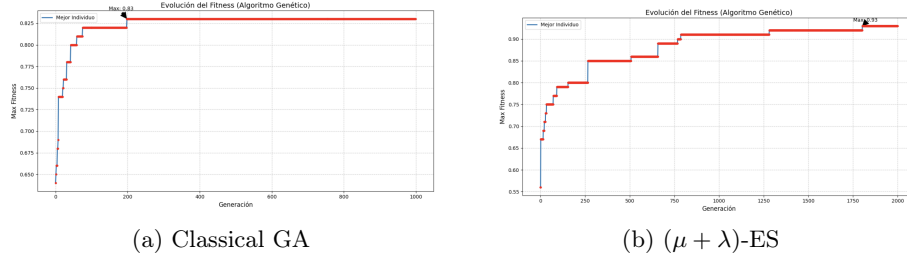


Fig. 2: Convergence of maximum fitness over generations in the French Flag Problem. (a) Classical GA stalls at a local optimum due to premature convergence. (b) $(\mu + \lambda)$ -ES strategy achieves progressive, staircase-shaped improvement.

$m(f)$ is defined as:

$$m(f) = \begin{cases} 25, & \text{if } 0 \leq f < 0.25 \\ 18, & \text{if } 0.25 \leq f < 0.45 \\ 12, & \text{if } 0.45 \leq f < 0.60 \\ 8, & \text{if } 0.60 \leq f < 0.75 \\ 4, & \text{if } 0.75 \leq f \leq 1 \end{cases} \quad (1)$$

where these empirically determined thresholds correspond to approximately 5%, 3.5%, 2.3%, 1.6%, and 0.8% of L , respectively. This ensures the schedule scales proportionally to the chromosome length and generalizes across problem sizes.

3.3 Evaluation Framework for 2D Morphogenesis

While a simple accuracy metric suffices for one-dimensional cases, evaluating two-dimensional morphogenesis requires capturing spatial, structural, and statistical nuances. Drawing inspiration from standard perceptual and statistical measures, we designed a composite framework integrating:

1. *Multichannel RGB Structural Similarity Index (SSIM)*. Given windows x and y , SSIM [17] evaluates similarity based on luminance, contrast, and structure. Since CA states are nominal, we map them into an RGB color space $([0, 255]^3)$ before computation to accurately penalize visual noise:

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)} \quad (2)$$

2. *State-wise Jaccard Index*. To rigorously evaluate spatial exactness, we average the Jaccard index [8] for each state $s \in \varepsilon$ independently, which equitably penalizes both false negatives and false positives:

$$\text{Jaccard}(X, Y) = \frac{1}{|\varepsilon|} \sum_{s \in \varepsilon} \frac{|X_s \cap Y_s|}{|X_s \cup Y_s|} \quad (3)$$

3. *Mutual Information* (MI). To handle cases where states might be inverted or translated, we incorporate MI [4] to measure the statistical dependence and reduction of uncertainty:

$$MI(X, Y) = H(X) + H(Y) - H(X, Y) \quad (4)$$

Finally, the global fitness function is defined as a convex combination of these three metrics, ensuring the final score remains bounded within $[0, 1]$:

$$f_{\text{fit}} = \alpha \cdot \text{SSIM} + \beta \cdot \text{Jaccard} + \gamma \cdot \text{MI} \quad (5)$$

where $\alpha + \beta + \gamma = 1$, and $\text{SSIM}, \text{Jaccard}, \text{MI} \in [0, 1]$.

4 CBR Engine

Configuring the CA and the evolutionary strategy from scratch for every new target pattern requires extensive manual testing and immense computational effort. While standard Machine Learning techniques could theoretically predict these parameters, they depend on massive training datasets and struggle to map a 2D image matrix to a complex multi-variable algorithmic configuration.

Each fully optimised CA rule encodes expert knowledge about how to generate a specific morphogenetic pattern. Rather than discarding this knowledge after solving each problem, the CBR engine stores it as a *case* and retrieves structurally similar instances. Crucially, in this work, new 2D target patterns are compared and retrieved exclusively against an initial 2D case base containing the foundational solved patterns of Japan and Hungary. This architecture follows the classic 4R CBR cycle: Retrieve, Reuse, Revise, Retain.

Each case $\mathcal{C}$ is formalized as a tuple:

$$\mathcal{C} = \langle P, \varphi^*, \Theta_{CA}, \Theta_{GA}, f^* \rangle, \quad (6)$$

where P is the problem descriptor (target pattern matrix, dimensionality, number of states, presence of horizontal/vertical stripes, number of connected components, uniform borders, and detected geometric patterns), φ^* is the best rule found, Θ_{CA} and Θ_{GA} are the CA and GA hyperparameter configurations, and f^* is the final fitness achieved.

4.1 Similarity Metric for Case Retrieval

When a new target pattern P_{new} is introduced, its topological features are extracted. The engine evaluates its similarity against the initial case base (Japan and Hungary) using a composite similarity function. The weights assigned to each similarity component were determined empirically through iterative testing. The dominant weight of $\text{Sim}_{\text{struct}}$ (0.5) reflects its role as the primary predictor of rule transferability, as structural topology directly determines neighborhood activation patterns. The lower weight of $\text{Sim}_{\text{states}}$ (0.1) is justified by the binary-to-binary compatibility check already embedded in the transfer learning phase.

$$Sim = 0.2 \cdot Sim_{topo} + 0.5 \cdot Sim_{struct} + 0.1 \cdot Sim_{states} + 0.2 \cdot Sim_{dims} \quad (7)$$

To ensure reproducibility, each component is strictly bounded within $[0, 1]$:

- i) Sim_{topo} computes the weighted average of boolean matches for topological features (e.g., stripes, borders).
- ii) Sim_{struct} averages three sub-metrics: Mutual Information (statistical dependence), the inverse Euclidean distance of state distribution histograms, and connected components similarity calculated as $1/(1 + |C_1 - C_2|)$.
- iii) Sim_{states} yields 1 if $|N_1 - N_2| = 0$, 0.5 if the difference is 1, and 0 otherwise.
- iv) Sim_{dims} returns 1 if both problems operate in the same dimensionality.

4.2 Reuse, Revision and Retention

Reuse (Transfer Learning). Based on similarity scores, implicit parameters are automated, qualitative settings are inherited from the best matched case within the initial memory, and quantitative ones (e.g., mutation, population size) are calculated as a weighted average. Crucially, if the retrieved case shares the same state space and neighborhood type, *Transfer Learning* is applied by injecting its best transition rule into the new GA’s initial population: 5% as exact clones and 5% as slightly mutated copies. This specific 10% ratio seeds high-quality genetic material while deliberately maintaining diversity; a larger proportion would risk flooding the population with overly similar individuals, reducing diversity and driving the search toward local optima.

Revision and Retention. The evolutionary framework runs using this automated parameterization and the injected seeds. If the best fitness f_{new}^* exceeds a dynamically adapting threshold θ_{retain} —which grows with the case base size (0.65, 0.75, and 0.85 for bases up to 10, 15, and beyond cases) to ensure only increasingly competitive solutions are stored— the new case is retained.

5 Experiments and Results

To populate the initial CBR case base and validate our evolutionary framework, we first evaluated classical 1D and 2D morphogenesis problems. Subsequently, the CBR engine was tested against a novel dataset of 12 patterns to measure its efficiency compared to random and heuristic approaches.

All experiments were executed on an Apple MacBook Air equipped with an Apple M1 chip (8-core CPU) and 8 GB of unified memory, running macOS. Table 1 details the fine-tuned parameters for the initial base cases stored in the CBR memory, alongside the generic Expert Heuristic baseline configuration that will be evaluated later as a control group.

Table 1: Hyperparameter configurations for the generic baseline heuristic and the specialized CBR base cases. Fitness weights are represented as a percentage tuple: (Accuracy, SSIM, Jaccard, MI).

Configuration	Neigh.	Pop.	Gen.	Steps (T)	Init.	Mutations	Weights (%)
Heuristic Baseline	Var.	250	500	60	Variable	Adapt. {15, 10, 6, 3}	(0, 35, 35, 30)
Drosophila (1D)	3	250	1000	90	Random Static	(3 genes)	(100, 0, 0, 0)
France (1D)	1	250	2000	90	Random Static	(1 gene)	(100, 0, 0, 0)
Hungary (2D)	Neumann	300	2000	75	Random Adapt.	{10, 8, 6, 4, 2}	(0, 80, 20, 0)
Japan (2D)	Moore	1000	2000	20	Seed	Adapt. {25, 18, 12, 8, 4}	(0, 20, 20, 60)

5.1 1D Morphogenesis Base Cases

Density Classification Problem (DCP): On the DCP benchmark [7], our classic GA was compared against the GKL rule across 250 random configurations ($N \in \{50, 75, 150\}$). While the GA achieved competitive accuracy (93.80% for $N = 50$, dropping to 91.40% for $N = 150$), the GKL rule exhibited scale invariance, scoring between 96.20% and 98.00%.

French Flag Problem: In the French Flag problem, the $\mu + \lambda$ strategy successfully overcame the premature convergence commonly observed in classic GAs, achieving a 95% fitness and demonstrating excellent generalisation across different automaton sizes (e.g., 30, 60, and 90 cells) to form the three distinct axial regions.

Drosophila Melanogaster Embryogenesis: As a systems biology proof-of-concept, the 1D framework modeled *Drosophila* embryogenesis. It successfully self-organised a random initial state into the biological 7-stripe pattern (70% fitness, 1000 gen.), demonstrating its ability to infer local rules from macroscopic phenotypes.

5.2 2D Morphogenesis Base Cases

Hungarian Flag (Horizontal Stripes): Relying solely on SSIM initially resulted in monochromatic blocks. However, employing our composite metric—combining SSIM (80%) with the state-wise Jaccard Index (20%)—alongside dynamic mutation allowed the automaton to escape local minima and form the flag.

Japanese Flag (Temporal Emergence): Growing a circle from a single central seed utilised a composite metric heavily weighted towards MI (60

5.3 CBR Evaluation and Performance Analysis

To evaluate its efficiency and generalization, the CBR engine was tested on 6 unseen 2D topological problems using exclusively the foundational memory (Japan and Hungary). Optimization performance was benchmarked across four distinct configurations: *Random*, *Expert Heuristic* (a domain-informed static

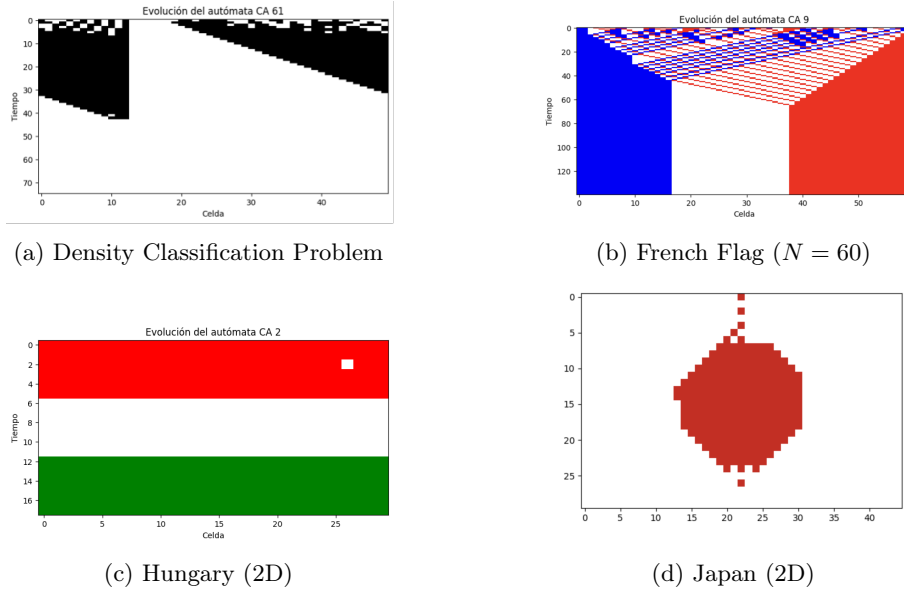


Fig. 3: Morphogenesis Base Cases: 1D spatiotemporal emergence (top) and 2D evolved patterns (bottom).

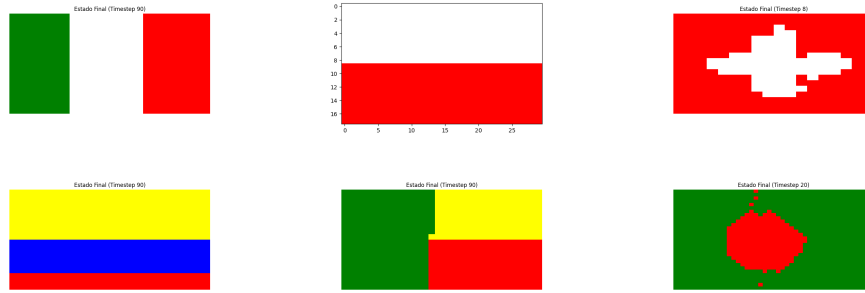


Fig. 4: Evolved 2D patterns successfully generated using the CBR Engine.

baseline detailed in Table 1), *CBR w/o TL* (an ablation variant lacking transfer learning), and the proposed *CBR engine*.

Beyond converging faster, the CBR-seeded evolutionary process demonstrated remarkable topological adaptation. For the Italian flag, the engine correctly matched the Hungarian flag from the initial case base, successfully driving the structural optimization from horizontal distributions toward vertical stripes. In the Bangladesh flag problem, the engine retrieved the Japan case and successfully adapted the injected rule to perform a spatial translation of the circular morphogenetic target.

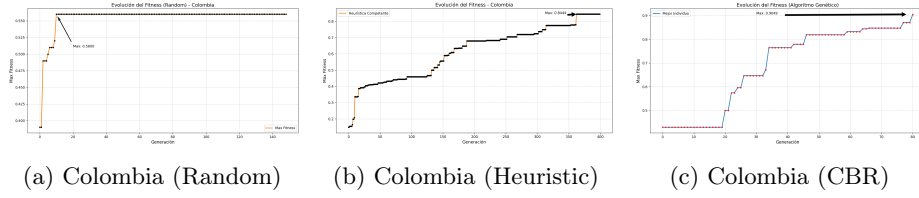


Fig. 5: Comparison of the maximum fitness evolution for the Colombia flag problem under the three studied scenarios: random search, heuristic configuration, and optimisation using the CBR engine.

To confirm the performance advantages discovered during the individual exploratory runs, a rigorous statistical evaluation was conducted. The framework was executed across $N = 5$ independent initialization seeds for every target case under four distinct computational conditions:

1. **Random:** A baseline where both the initial population and optimization setup (e.g., fitness metric weights, generational limits) are selected entirely at random without domain guidance. Consequently, some high fitness occasionally achieved under this condition may stem from lucky parameter combinations or miscalibrated objective weighting rather than structured convergence towards the desired target morphology.
2. **Expert Heuristic:** The generic static configuration baseline using pre-engineered, fixed parameters across all cases.
3. **CBR w/o TL:** An ablation variant where the CBR engine automates hyperparameter selection but transfer learning (rule seeding) is disabled, isolating the contribution of parameter tuning from initial genetic knowledge injection.
4. **CBR (proposed):** The full proposed engine seeded exclusively with the foundational case memory of Japan and Hungary.

Table 2 aggregates the quantitative results of this analysis, reporting the mean and standard deviation of both the final achieved fitness and the total generations required to satisfy the convergence or stagnation stopping criteria. Additionally, to verify whether the differences in optimization quality are statistically meaningful across all multi-seed executions, a non-parametric pooled paired Wilcoxon signed-rank test was computed. The resulting significance levels and empirical effect sizes are detailed in Table 3. Finally, the aggregate spatiotemporal performance profile and learning trajectory for each target case and experimental condition are graphically visualized in Fig. 6.

Table 2: Statistical validation across $N = 5$ independent seeds per case/condition. Values report mean $\pm$ standard deviation of final fitness, generations to stop (convergence to $f \geq 0.9$ or 100-generation stagnation), and convergence rate (% of seeds reaching $f \geq 0.9$). **CBR (proposed)** denotes the full engine operating exclusively over the Japan and Hungary base configurations.

Case	Condition	Fitness (mean $\pm$ std)	Generations (mean $\pm$ std)	Convergence rate
Italy	Random	0.816 ± 0.105	150.8 ± 87.7	0%
	Heuristic	0.713 ± 0.114	378.4 ± 48.3	0%
	CBR w/o TL	0.589 ± 0.204	560.6 ± 267.5	0%
	CBR (proposed)	0.908 ± 0.006	132.2 ± 23.9	100%
Poland	Random	0.832 ± 0.072	49.2 ± 74.0	20%
	Heuristic	0.915 ± 0.011	143.2 ± 68.7	100%
	CBR w/o TL	0.904 ± 0.004	192.0 ± 36.7	100%
	CBR (proposed)	0.908 ± 0.011	132.6 ± 73.3	100%
Benin	Random	0.715 ± 0.130	241.8 ± 136.5	0%
	Heuristic	0.810 ± 0.098	385.8 ± 20.8	40%
	CBR w/o TL	0.883 ± 0.044	537.6 ± 187.4	80%
	CBR (proposed)	0.902 ± 0.002	142.2 ± 9.6	100%
Swiss	Random	0.702 ± 0.264	104.4 ± 82.5	40%
	Heuristic	0.476 ± 0.036	290.0 ± 67.5	0%
	CBR w/o TL	0.830 ± 0.026	500.6 ± 103.1	0%
	CBR (proposed)	0.872 ± 0.013	395.4 ± 43.8	0%
Bangladesh	Random	0.688 ± 0.221	77.6 ± 83.6	20%
	Heuristic	0.546 ± 0.041	286.2 ± 30.5	0%
	CBR w/o TL	0.750 ± 0.070	579.0 ± 152.2	0%
	CBR (proposed)	0.885 ± 0.038	612.4 ± 113.9	40%
Colombia	Random	0.770 ± 0.098	233.8 ± 156.9	20%
	Heuristic	0.788 ± 0.062	400.0 ± 0.0	0%
	CBR w/o TL	0.774 ± 0.072	964.8 ± 195.2	0%
	CBR (proposed)	0.817 ± 0.020	367.2 ± 41.6	0%

Table 3: Pooled paired Wilcoxon signed-rank tests (paired by seed across the 6 evaluated cases) comparing the proposed CBR engine against each baseline/ablation on final fitness. r is the matched-pairs rank-biserial correlation (effect size, $r \in [-1, 1]$); n is the number of paired observations.

Comparison (CBR vs.)	n	Mean CBR	Mean baseline	p -value	Effect size r
Heuristic	30	0.882	0.708	< 0.001 ***	0.849
Random	30	0.882	0.754	< 0.001 ***	0.763
CBR w/o Transfer Learning	30	0.882	0.788	< 0.001 ***	0.841

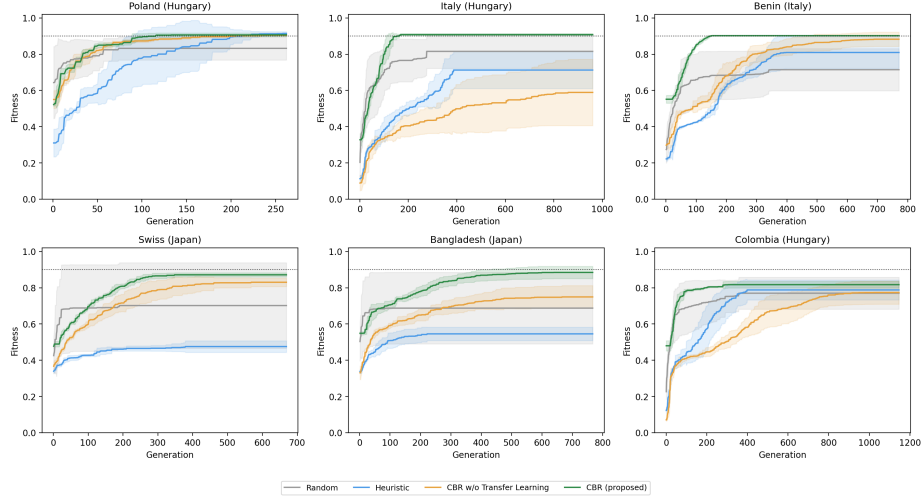


Fig. 6: Mean fitness evolution (± 1 standard deviation band, $N = 5$ seeds) per case and condition. Curves are forward-filled after each run's stopping generation (fitness is monotonically non-decreasing under $(\mu + \lambda)$ elitist selection), so the flat tail of a curve reflects the point at which that condition stopped (either by reaching $f \geq 0.9$ or by 100-generation stagnation), not a resumed search. The dotted horizontal line marks the $f = 0.9$ convergence threshold.

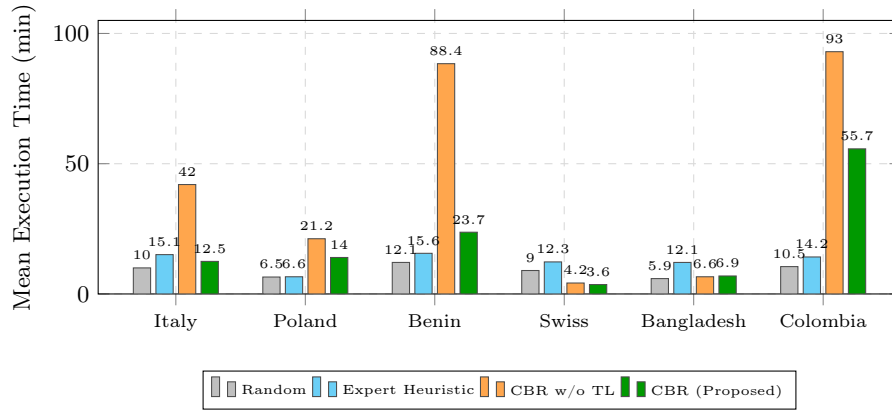


Fig. 7: Mean execution time in minutes per target case across experimental conditions ($N = 5$ independent runs per condition).

Computational Efficiency and Runtime Dynamics. Analyzing execution times (Fig. 7) highlights three main performance insights:

- **Transfer Learning Efficiency:** Rule seeding in *CBR (proposed)* reduces mean execution time by 54.4% compared to *CBR w/o TL* (19.4 min vs. 42.6 min). This impact is highest in complex 3-state topologies like *Benin* (23.7 min vs. 88.4 min, a 73.2% reduction) and *Colombia* (55.7 min vs. 93.0 min).
- **Superior Quality in Equal or Less Time:** In patterns such as *Swiss*, *Bangladesh*, and *Italy*, *CBR* achieves significantly higher fitness ($f \geq 0.87$) in less time than the *Expert Heuristic* (e.g., 3.6 min vs. 12.3 min in *Swiss*), which traps early at poor scores ($f \leq 0.71$).
- **Productive Search vs. Premature Stagnation:** Lower runtimes in baseline approaches (*Random* and *Expert Heuristic*) stem from premature stagnation in sub-optimal local minima ($f \approx 0.48$ – 0.78), triggering early stopping criteria. Conversely, additional runtime in *CBR* represents active, productive exploration driving true spatiotemporal convergence ($f \geq 0.82$ – 0.91).

6 Discussion and Conclusions

This work demonstrates that integrating evolutionary strategies and CBR provides a robust framework for optimizing CA rules in morphogenesis. The $(\mu + \lambda)$ -ES strategy outperforms classical GAs via adaptive mutation, while the composite fitness framework (SSIM, Jaccard, MI) guides evolution toward structurally sound patterns. Crucially, the CBR engine substantially improves computational efficiency through targeted transfer learning from the foundational memory (Japan and Hungary). Overall, rule seeding reduces average execution time by 54.4% (from 42.6 min down to 19.4 min per run) and cuts required generations by up to 76.4% for Italy, 73.5% for Benin, and 61.9% for Colombia while attaining significantly higher mean fitness.

Limitations and Future Work. While effective for clustered and axial patterns, deterministic discrete CAs struggle with: (i) perfectly symmetrical shapes (e.g., Austria or Czech flags), which require symmetry-breaking directional bias, and (ii) continuous animal textures (e.g., leopard coats).

Hence, future work will focus on three main pillars: (i) automating case-base expansion by systematically generating random target patterns to enrich the case base, (ii) exploring Neural Cellular Automata (NCA) to leverage continuous states and convolutional networks for complex textures, and (iii) moving beyond synthetic flag benchmarks toward real biological phenomena, such as tumor growth or wound healing [15], to provide a stronger validation of the framework.

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